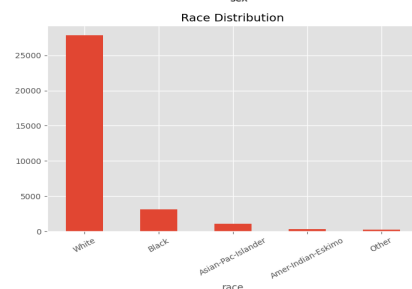
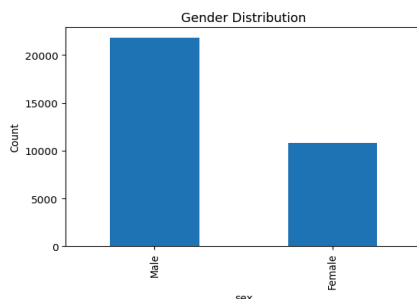
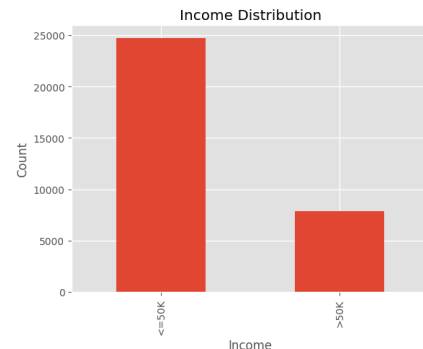


Abstract : The Adult Income Dataset from the UCI Machine Learning Repository is evaluated for likely biases in this work. The dataset's socioeconomic and demographic attributes are utilized to estimate whether a person's yearly income will approach \$50,000. Using Python, Pandas, NumPy, and Matplotlib, exploratory data analysis (EDA) was carried out to look at the distributions of gender, race, education, age, and income. The investigation found differences in income across educational levels and demographic imbalances, particularly in gender and race. These results show that the dataset displays historical and representation biases, which might have an affect on machine learning models that were trained using this data. Building equitable and responsible AI systems demands a grasp of these biases.

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2   fnlwgt               32561 non-null  int64
3   education            32561 non-null  object
4   education.num        32561 non-null  int64
5   marital.status       32561 non-null  object
6   occupation           32561 non-null  object
7   relationship         32561 non-null  object
8   race                 32561 non-null  object
9   sex                  32561 non-null  object
10  capital.gain          32561 non-null  int64
11  capital.loss          32561 non-null  int64
12  hours.per.week        32561 non-null  int64
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14  income               32561 non-null  object
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1. Overview : Patterns are immediately identified from data utilizing machine learning techniques. Unfair predictions may be created by the resultant models if there are demographic differences in the training data. Therefore, detecting bias prior to model creation is crucial to responsible AI. This project's objective is to employ exploratory data analysis to analyze the Adult Income Dataset to uncover any biases relating to gender, race, age, education, and income.

2. Description of the Dataset : 32,561 items and 15 variables, including demographic data like age, gender, race, education, profession, and weekly working hours, are included in the Adult Income Dataset. People are sorted into two income categories based on the target variable: $\leq 50K$ and $> 50K$. The dataset summary reveals that folks are 38.6 years old on average and work 40.4 hours a week on average shown in Figure 1 the Overview of the Dataset.

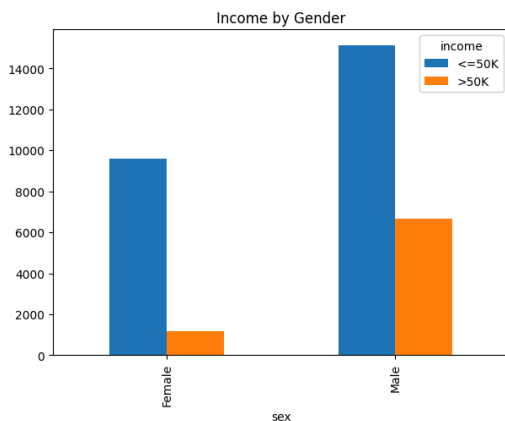
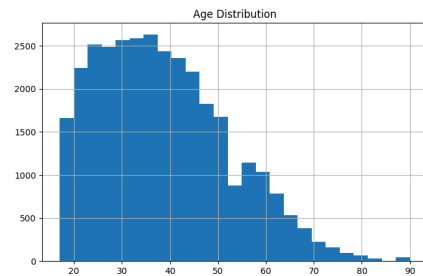
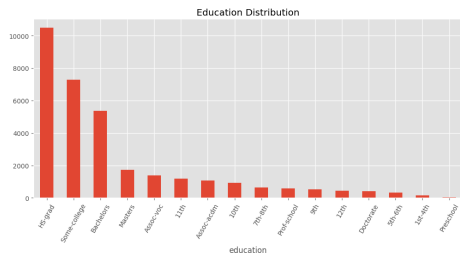
3. Techniques : Python was applied for the inquiry, along with the Matplotlib, NumPy, and Pandas packages. The dataset was loaded, missing values were verified, and then demographic features were exploratorily explored. Potential biases pertaining to gender, ethnicity, education, age, and income were found employing bar charts, histograms, and correlation analysis as shown in Figure 2 the Distribution of Income.

4.1 Gender Distribution Results : There are more males than women in the sample, which demonstrates a gender imbalance that might impact prediction models as shown in Figure 3 the Distribution of Gender.

4.2 Distribution of Races : White people made up 85.43% of the sample, followed by Black people (9.59%), Asian-Pacific Islanders (3.19%), Amer-Indian-Eskimo (0.96%), and Other (0.83%). This shows a substantial bias in representation as shown in Figure 4 the Distribution of Races

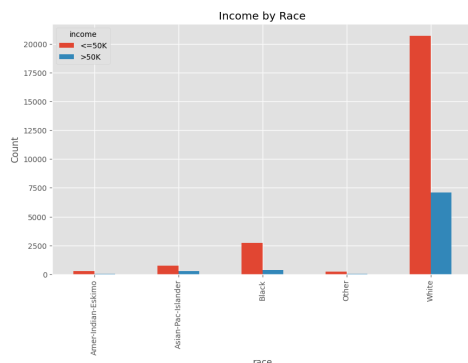
Examining Adult Income Dataset Bias

4.3 Distribution of Age and Education : The majority of folks are between the ages of 28 and 48, and the most prevalent educational levels are some college and high school completion. Advanced degrees are less frequent as shown in Figure 5 the Distribution of Ages and Figure 6 the Distribution of Education.

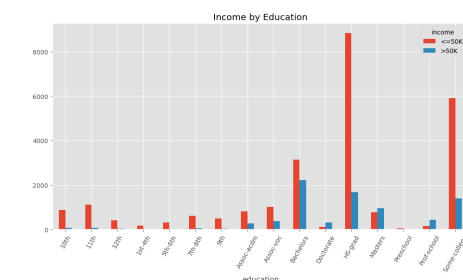


4.4 Findings Income by Gender : Just 10.95% of women earn more than \$50K, compared to 89.05% who make less than \$50K. 69.43% of males make less than \$50,000, while 30.57% make more than \$50,000. This clearly indicates that the distribution of income differs by gender as shown in Figure 7 the Gender-specific Income.

4.5 Income by Race : Because it makes up the majority of the information, the White group has the most individuals in both income groups. High-income results are much lower for minority populations as shown in Figure 8 the Income by Race.



4.6 Earnings by Education : The percentage of those with bachelor's, master's, professional school, and PhD degrees who earn more than \$50,000 is larger than that of persons with less education, suggesting that education has a substantial effect on income as shown in Figure 9 the Education and Income



5. Biases Found : Several kinds of bias were discovered in the analysis:

5.1 Gender Bias: Men are more likely than women to earn more than \$50,000.

5.2 Race Bias: Minority groups are underrepresented in the dataset, while white folks triumph.

5.3 Education Bias: Higher income is related with higher levels of education.

5.4 Representation Bias: There is an uneven distribution of the dataset across demographic groups.

5.5 Historical Bias: The 1994 U.S. Census's socioeconomic circumstances are reflected in the data.

Examining Adult Income Dataset Bias

6. Conclusion : There are clear socioeconomic and demographic biases in the Adult Income Dataset. The distribution of income is varied by gender, race, and education, and some groups are conspicuously underrepresented. These results underscore how crucial it is to examine datasets for fairness before utilizing them to train machine learning algorithms. Reducing prejudice and promoting the building of moral AI systems may be achieved by adopting fairness-aware tactics and getting more representative data.

7. References

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